Management Science Assignment 1 Solutions

Worked solutions for all nine problems in the revised assignment. Monetary values are expressed in the units specified in each problem.

Problem 1

Let x_1 and x_2 denote daily units of Products 1 and 2. Both variables are continuous and nonnegative.

Maximize $Z = 4x_1 + 3x_2$ (unit: KRW 10,000)

subject to

$2x_1 + x_2 \le 100$ (machine-hours)

$x_1 + x_2 \le 80$ (labor-hours)

$x_1, x_2 \ge 0$

The objective adds the profit from both products. The two resource constraints limit daily machine and labor use. There are no minimum-production or demand constraints.

Instructor check: the optimum is $x_1 = 20$ and $x_2 = 60$, with $Z = 260$, or KRW 2,600,000 per day. Students are asked only to formulate the model.

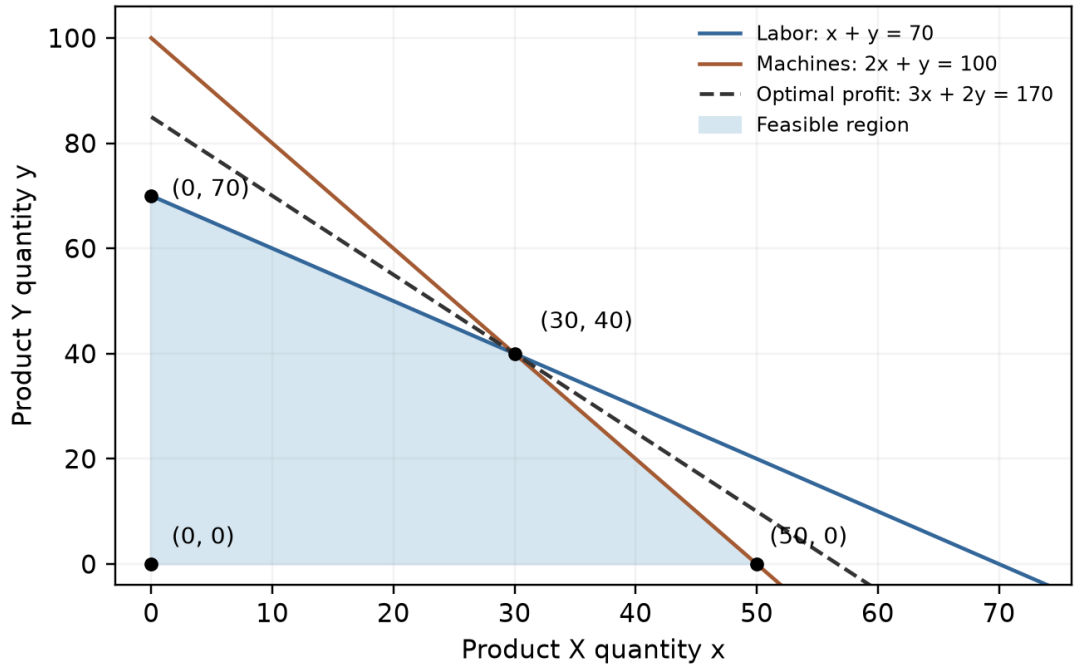
Problem 2

The boundary lines have intercepts (70, 0) and (0, 70) for labor, and (50, 0) and (0, 100) for machines. The feasible region is the area in the first quadrant below both lines.

$x + y = 70$; $2x + y = 100$ $\Rightarrow$ $x = 30, y = 40$.

Feasible vertex	Profit 3x + 2y
(0, 0)	0
(50, 0)	150
(30, 40)	170
(0, 70)	140

The unique optimal solution is (x, y) = (30, 40). Maximum weekly profit is 170 × KRW 10,000 = KRW 1,700,000. The objective slope −3/2 lies strictly between the two binding boundary slopes, −2 and −1, so no feasible edge has the same objective value.



Problem 3

(1) Resource slack equals capacity minus use. At s = 100 and h = 40, the calculations and (2) constraint classifications are:

Assembly: $180 - (100 + 2 \times 40) = 0$ hours → binding.

Testing: $140 - (100 + 40) = 0$ hours → binding.

Chips: $45 - 40 = 5$ chips → nonbinding.

(3) Assembly and testing capacity are fully used, while five chips remain unused. Increasing assembly or testing capacity can raise optimal profit within the relevant range; buying more chips alone would not help. Managers should compare the value of extra capacity with its cost before expanding.

Problem 4

```
import cvxpy as cp

# 1. Decision variables
s = cp.Variable()
h = cp.Variable()

# 2. Objective: daily profit in KRW 10,000
objective = cp.Maximize(3 * s + 5 * h)

# 3. Resource and nonnegativity constraints
constraints = [
    s + 2 * h <= 180,
    s + h <= 140,
    h <= 45,
    s >= 0,
    h >= 0
]

# 4. Formulate and solve
problem = cp.Problem(objective, constraints)
problem.solve()

# 5. Print results
print("Status:", problem.status)
print("Maximum profit (KRW 10,000):", problem.value)
print("Optimal s:", s.value)
print("Optimal h:", h.value)
```

Problem 5

Model A is infeasible. The lower bounds imply $x + y \geq 4 + 3 = 7$, contradicting $x + y \leq 5$. No feasible solution exists.

Model B is unbounded above. Set $x = y = t$ for any $t \geq 0$. Then $x - y = 0 \leq 4$ and the objective is $2t + t = 3t$, which tends to infinity as t increases.

The student's claim is false. Infeasibility means that no point satisfies all constraints. Unboundedness means feasible points exist, but the objective can improve without a finite upper bound.

Problem 6

The modified objective is $cx + 2y$, so its level lines are $y = Z/2 - (c/2)x$. Their slope is $-c/2$. The labor and machine boundary slopes are -1 and -2.

$-2 \leq -c/2 \leq -1 \quad \Rightarrow \quad 2 \leq c \leq 4.$

Thus (30, 40) remains optimal for $2 \leq c \leq 4$, and is uniquely optimal for $2 < c < 4$. At $c = 2$, every point on the edge from (0, 70) to (30, 40) is optimal. At $c = 4$, every point on the edge from (30, 40) to (50, 0) is optimal.

For $c < 2$, moving toward (0, 70) increases profit; for $c > 4$, moving toward (50, 0) increases profit. The original coefficient $c = 3$ lies strictly inside the allowable range.

Problem 7

(1) Each shadow price is the change in maximum profit from one additional resource hour. The table reports profit in units of KRW 10,000:

Mixing: (2,900 − 2,900) × 10,000 / 1 = KRW 0 per hour.

Molding: (2,902 − 2,900) × 10,000 / 1 = KRW 20,000 per hour.

Testing: (2,903 − 2,900) × 10,000 / 1 = KRW 30,000 per hour.

(2) Mixing has a shadow price of zero: a small capacity increase adds no profit because the original plan uses only 700 of the available 800 mixing hours.

One additional molding hour raises maximum monthly profit by KRW 20,000; one additional testing hour raises it by KRW 30,000, before any capacity-acquisition cost. Thus the marginal value per hour ranks testing first, molding second, and mixing last. An investment decision must also consider the cost of obtaining extra capacity.

These are marginal values within the applicable allowable ranges, with other data fixed. They do not imply that unlimited capacity expansion is profitable.

Problem 8

(1) Requiring one ton of D changes maximum profit from 2,900 to 2,898 units of KRW 10,000. Using the maximization sign convention, its reduced cost is:

Reduced cost = (2,898 − 2,900) × 10,000 / 1 = −KRW 20,000 per ton.

(2) Producing one ton of D reduces the best achievable profit by KRW 20,000 after the other quantities are reoptimized. Although D earns a positive unit profit, its resources earn more in the alternative production mix. This is why D is not produced in the original optimal plan.

With all other data unchanged, D may be produced when its unit profit rises by KRW 20,000 to KRW 120,000 per ton: it then has zero reduced cost and can appear in an equally profitable optimal plan. Above KRW 120,000, introducing D improves profit relative to the original plan. Changes in resource requirements or availability could also make D attractive, but would require reoptimization.

Resource-value check: 3(0) + 3(20,000) + 2(30,000) = KRW 120,000 per ton of D. Therefore reduced cost = 100,000 − 120,000 = −KRW 20,000 per ton, consistent with the table.

Problem 9

(1) Each outsourced molding hour costs KRW 30,000 but avoids KRW 20,000 of in-house cost already deducted from the original unit profits.

Net additional cost = 30,000 − 20,000 = KRW 10,000 per hour.

Marginal net benefit = 20,000 − 10,000 = KRW 10,000 per hour.

(2) Additional hours within the current shadow-price range = 750 − 700 = 50 hours.

Recommended outsourcing = min(120, 50) = 50 molding hours.

Total profit gain = 50 × 10,000 = KRW 500,000.

Net monthly profit increases from KRW 29,000,000 to KRW 29,500,000. Each of the first 50 outsourced hours adds KRW 20,000 in operating profit and costs a net KRW 10,000, so each contributes a net gain of KRW 10,000.

Operating-profit upper bound (unit: KRW 10,000): $4A + 8B + 5C + 10D \le 6(A + B + C + 2D) \le 6(500) = 3{,}000$. The first inequality follows from $A \ge B$ and $C, D \ge 0$.

Check beyond 50 hours: the original shadow price cannot be extrapolated above 750 molding hours. At that capacity, $A = B = 250$ and $C = D = 0$ yields KRW 30,000,000 before net outsourcing costs and fully uses testing capacity. No production mix can exceed this operating profit under the existing testing capacity and $A \ge B$ policy. Additional molding hours therefore add no operating profit, while their net cost remains KRW 10,000 per hour. Thus 50 hours is also the best total purchase.